

$$= \frac{2}{2 + \frac{2}{(39/11)} \times 0.39}$$

$$= \frac{2}{2 + \frac{22}{39} \times \frac{39}{100}} = \frac{2}{2 + \frac{22}{100}}$$

$$= \frac{2}{2 + \frac{11}{50}} = \frac{2}{(111/50)} = \frac{100}{111}.$$

168. Given exp. = $\frac{1}{1 + \frac{\frac{2}{3}}{\frac{5}{3} + \frac{8}{9} \times 3}} = \frac{1}{1 + \frac{2/3}{5/3 + 8/3 \times 3}}$

$$= \frac{1}{1 + \frac{2/3}{(13/3)}} = \frac{1}{1 + \frac{2}{13}} = \frac{13}{15}.$$

169. $\frac{2}{2 + \frac{2}{2 + \frac{2}{x}}} = 3 \Leftrightarrow \frac{2}{2 + \frac{2}{\left(\frac{2x+2}{x}\right)}} = 3$

$$\Leftrightarrow \frac{2}{2 + \frac{2x}{2x+2}} = 3 \Leftrightarrow \frac{2}{\frac{2(2x+2)+2x}{2x+2}} = 3$$

$$\Leftrightarrow \frac{2(2x+2)}{2(3x+2)} = 3$$

$$\Leftrightarrow \frac{(2x+2)}{(3x+2)} = 3 \Leftrightarrow 2x+2 = 9x+6$$

$$\Leftrightarrow 7x = -4 \Leftrightarrow x = -\frac{4}{7}.$$

170. $2 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}} = \frac{37}{13} = 2\frac{11}{13} = 2 + \frac{11}{13} \Rightarrow \frac{1}{x + \frac{1}{y + \frac{1}{z}}} = \frac{11}{13}$

$$= \frac{11}{13} \Rightarrow x + \frac{1}{y + \frac{1}{z}} = \frac{13}{11}$$

$$\Rightarrow x + \frac{1}{y + \frac{1}{z}} = 1 + \frac{2}{11}$$

$$\Rightarrow x = 1, y + \frac{1}{z} = \frac{11}{2} = 5\frac{1}{2} = 5 + \frac{1}{2}$$

$$\Rightarrow x = 1, y = 5, z = 2.$$

171. Clearly, we have: $x = 1 + \frac{1}{x} = \frac{x+1}{x}.$

$$\Rightarrow x^2 = x + 1 \Rightarrow x^2 - x - 1 = 0.$$

172. Let $x = \frac{1}{3 + \frac{1}{4 + \dots}}$

$$\dots$$

$$+ \frac{1}{1008}$$

Then, $P = \frac{1}{1 + \frac{1}{1+x}} = \frac{1+x}{2+x}, Q = \frac{1}{2+x}.$

$$\therefore P + Q = \frac{1+x}{2+x} + \frac{1}{2+x} = \frac{2+x}{2+x} = 1.$$

173. Given exp. = $\frac{a^2 - b^2}{a+b}$, where $a = \left(1 + \frac{1}{10 + \frac{1}{10}}\right)$ and

$$b = \left(1 - \frac{1}{10 + \frac{1}{10}}\right)$$

$$= (a-b) = \left(1 + \frac{1}{10 + \frac{1}{10}}\right) - \left(1 - \frac{1}{10 + \frac{1}{10}}\right)$$

$$= 2 \times \frac{1}{(101/10)} = \frac{20}{101}.$$

174. $3x + y = 19 \dots(i)$ and $x - y = 9 \dots(ii)$

Adding (i) and (ii), we get: $4x = 28$ or $x = 7.$

Putting $x = 7$ in (i), we get: $y = -2.$

175. $4x = p(x+3) + q(x-1) \Rightarrow 4x = px + 3p + qx - q$

$$\Rightarrow 4x = (p+q)x + (3p-q)$$

$$\Rightarrow p+q = 4 \text{ and } 3p-q = 0$$

$$\Rightarrow p+q = 4 \text{ and } p = \frac{q}{3}$$

$$\Rightarrow \frac{q}{3} + q = 4 \Rightarrow \frac{4q}{3} = 4 \Rightarrow q = 3.$$

$$p+q = 4 \text{ and } q = 3 \Rightarrow p = 1.$$

176. $xyz = 256 \Rightarrow (2z)(2z)z = 256 \Rightarrow 4z^3$

$$= 256 \Rightarrow z^3 = 64 \Rightarrow z = 4.$$

$$\therefore x = 2z = (2 \times 4) = 8.$$

177. $\frac{28x}{13y} = \frac{140}{39} \Rightarrow \frac{x}{y} = \left(\frac{140}{39} \times \frac{13}{28}\right) = \frac{5}{3} \Rightarrow x = \frac{5}{3}y.$

$$3y + 9x = 54 \Rightarrow y + 3x = 18$$

$$\Rightarrow y + 3 \times \frac{5}{3}y = 18 \Rightarrow 6y = 18 \Rightarrow y = 3.$$

$$\therefore x = \left(\frac{5}{3} \times 3\right) = 5.$$

So, $y - x = (3 - 5) = -2.$

178. $3x + 7y = 75 \dots(i)$

$$5x - 5y = 25 \Rightarrow x - y = 5 \Rightarrow 7x - 7y = 35 \dots(ii)$$

Adding (i) and (ii), we get: $10x = 110$ or $x = 11.$

Putting $x = 11$ in (i), we get: $7y = 42$ or $y = 6.$

$$\therefore x + y = (11 + 6) = 17.$$

179. $a + b = 5 \dots(i)$ and $3a + 2b = 20 \dots(ii)$

Multiplying (i) by 2 and subtracting from (ii), we get: $a = 10.$

Putting $a = 10$ in (i), we get: $b = -5.$

$$\therefore (3a + b) = 3 \times 10 + (-5) = 30 - 5 = 25.$$

180. $(2p + 3q) + (2p - q) = 18 + 2 \Rightarrow 4p + 2q = 20$

$$\Rightarrow 2(2p + q) = 20 \Rightarrow 2p + q = 10.$$